

TAMIL NADU STATE BOARD - STANDARD 9 SCIENCE (PHYSICS)

CHAPTER 1: PHYSICS

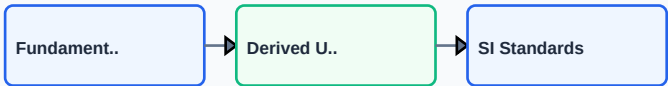
PREMIUM ACADEMIC STUDY SUITE & EXAM GUIDE

Board:	Tamil Nadu State Board (TNSB)
Syllabus:	Samacheer Kalvi Textbook Curriculum aligned
Standard:	Class 9 English Medium
Subject:	Science (Physics)
Chapter Name:	Physics

Learning Objectives:

- Comprehend core theoretical concepts and exact textbook terminology.
- Apply relevant formulas and proofs to solve high-scoring board problems.
- Interpret and sketch vector diagrams or graphical models representing equations.
- Build examination readiness for 1-mark, 2-mark, and 5-mark question sets.

TOPIC ROADMAP: MEASUREMENT STANDARDS



Core Concepts & Definitions

Textbook definitions are essential for scoring fully in the short-answer section of board assessments. Review the primary definitions below:

Term	Official Definition & Context
Physical Quantity	A quantity that can be measured and in terms of which laws of physics can be described, consisting of a numerical value and a unit.
Fundamental Quantities	Quantities that cannot be expressed in terms of any other physical quantities, such as length, mass, time, temperature, electric current, luminous intensity, and amount of substance.
Derived Quantities	Quantities that can be expressed in terms of fundamental quantities, like area, volume, velocity, acceleration, force, and pressure.
Unit	The standard measure of a physical quantity used for comparison, such as metre, kilogram, or second.
Fundamental Units	The units of fundamental quantities, defined independently (e.g., metre for length, kilogram for mass, second for time).
Derived Units	The units used to measure derived quantities, expressed as combinations of fundamental units (e.g., m/s for velocity, kg/m ³ for density).
Metre	The SI base unit of length, defined as the distance travelled by light in vacuum during a time interval of $1/299,792,458$ of a second.
Kilogram	The SI base unit of mass, defined by taking the fixed numerical value of the Planck constant h to be $6.62607015 \times 10^{-34}$ when expressed in J s.
Second	The SI base unit of time, defined by taking the fixed numerical value of the caesium frequency $\Delta\nu_{\text{Cs}}$ to be 9,192,631,770 when expressed in Hz.
Kelvin	The SI base unit of thermodynamic temperature, defined by taking the fixed numerical value of the Boltzmann constant k to be 1.380649×10^{-23} when expressed in J K ⁻¹ .
Astronomical Unit	The mean distance between the center of the Earth and the center of the Sun, equal to 1.496×10^{11} metres.
Light Year	The distance travelled by light in vacuum in one year, equal to 9.46×10^{15} metres.
Parsec	Parallactic second, defined as the distance at which an arc of length 1 astronomical unit subtends an angle of one second of arc, equal to 3.08×10^{16} metres.
Least Count	The smallest value that can be measured accurately with a given measuring instrument (e.g., 0.1 mm for Vernier calipers, 0.01 mm for screw gauges).
Zero Error	The error in a measuring instrument when the zero mark of the measuring scale does not coincide with the reference zero index line under no-load conditions.
Pitch	The distance travelled by the spindle of a screw gauge per one complete rotation of the thimble / head scale.

Vernier Scale	A small, auxiliary graduated scale that slides along the main scale of an instrument, allowing readings to a fraction of a division.
Screw Gauge	A high-precision instrument used to measure dimensions in the millimetre range, such as wire diameter or metal sheet thickness, using a fine pitch screw.
Spring Balance	A device used to measure the weight of an object based on Hooke's Law, which states that the stretch of a spring is proportional to the force applied.

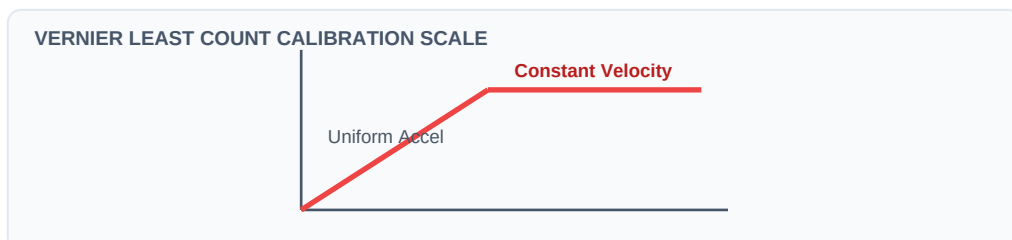
Syllabus Exam Tip: Memorize precise keywords. Examiners award maximum marks when textbook terms and correct symbols are correctly referenced.

GLOSSARY: SI BASE UNITS STANDARD FLOW



Detailed Explanation & Formulas

Here, we analyze the core laws and formulas. Examine the conceptual illustration below to visual the relation between variables:



Essential Formulas, Laws & Identities:

- Least Count of Vernier Caliper (LC) = 1 Main Scale Division (MSD) - 1 Vernier Scale Division (VSD) = 0.1 mm = 0.01 cm
- Vernier Total Reading (TR) = Main Scale Reading (MSR) + (Vernier Coincidence (VC) * LC) ± Zero Correction
- Zero Correction (ZC) = - (Zero Error)
- Least Count of Screw Gauge (LC) = Pitch / Number of Head Scale Divisions = 1 mm / 100 = 0.01 mm = 0.001 cm
- Pitch of Screw Gauge = Distance moved on Pitch Scale / Number of complete rotations
- Screw Gauge Total Reading (TR) = Pitch Scale Reading (PSR) + (Head Scale Coincidence (HSC) * LC) ± Zero Correction
- Weight of a body (W) = Mass (m) * Acceleration due to gravity (g) | $W = m * g$ (where $g \approx 9.8 \text{ m/s}^2$ on Earth)
- Astronomical Unit (1 AU) = $1.496 \times 10^{11} \text{ m}$
- Light Year (1 ly) = $9.46 \times 10^{15} \text{ m}$
- Parsec (1 pc) = $3.08 \times 10^{16} \text{ m}$
- Density (ρ) = Mass (m) / Volume (V)

Make sure to check the dimensional consistency of all physical quantities and verify that units are correctly stated in final solutions.

Worked Examples

Review these standard board examination-style worked examples to learn proper step layout structures:

Example 1: A Vernier caliper has a main scale reading of 4.5 cm and a Vernier coincidence of 6. If the positive zero error is 0.03 cm, calculate the corrected reading.

Step-by-step Solution:

Given: MSR = 4.5 cm VC = 6 LC = 0.01 cm Zero Error = +0.03 cm Zero Correction (ZC) = -0.03 cm Formula: TR = MSR + (VC * LC) + ZC Calculation: TR = 4.5 + (6 * 0.01) - 0.03 TR = 4.5 + 0.06 - 0.03 TR = 4.53 cm.

Example 2: A screw gauge has a pitch of 1 mm and 100 divisions on its head scale. The thimble shows a pitch scale reading of 3 mm and head scale coincidence of 45. Find the corrected reading if the instrument has a negative zero error of 0.04 mm.

Step-by-step Solution:

Given: Pitch = 1 mm Head scale divisions = 100 LC = Pitch / 100 = 0.01 mm PSR = 3 mm HSC = 45 Zero Error = -0.04 mm Zero Correction (ZC) = +0.04 mm Formula: TR = PSR + (HSC * LC) + ZC Calculation: TR = 3 mm + (45 * 0.01 mm) + 0.04 mm TR = 3 + 0.45 + 0.04 TR = 3.49 mm.

Example 3: Convert an astronomical distance of 2.5 light years into standard SI units (meters).

Step-by-step Solution:

Given: Distance = 2.5 light years (ly) 1 light year = 9.46×10^{15} m Formula: Distance in meters = Distance in ly * (9.46×10^{15} m/ly) Calculation: Distance = 2.5 * 9.46×10^{15} Distance = 23.65×10^{15} m = 2.365×10^{16} m.

WORKED: READING VERNIER CALIPERS DIMENSION



HOTS (Higher Order Thinking Skills) Challenge

Challenge yourself with these complex, multi-concept problems designed to test deeper conceptual understanding and mathematical reasoning:

HOTS Challenge 1: Why are astronomical distances expressed in light years or parsecs instead of standard SI units like meters?

Analytical Solution Strategy:

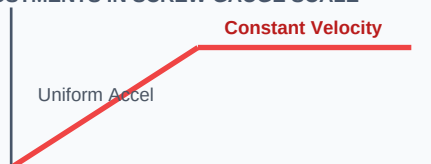
Astronomical distances (e.g. between stars and galaxies) are incredibly large. Writing them in meters yields extremely large numbers (e.g. Earth to Sun is 149,600,000,000 meters) which are prone to clerical and calculation errors. Light years (distance traveled by light in a year) and parsecs simplify these numbers to manageable quantities.

HOTS Challenge 2: A spring balance calibrated on Earth is taken to the Moon to measure the mass and weight of a rock. Explain how the readings of the spring balance will change.

Analytical Solution Strategy:

A spring balance measures weight (gravitational force, $W = m \cdot g$) rather than mass. Since acceleration due to gravity on the Moon ($g_{\text{moon}} \approx 1.63 \text{ m/s}^2$) is 1/6th of that on Earth ($g_{\text{earth}} \approx 9.8 \text{ m/s}^2$), the weight reading on the spring balance will decrease to 1/6th of its Earth value. The rock's actual mass remains unchanged, but a spring balance cannot measure mass directly without recalibration.

HOTS: ZERO ERROR ADJUSTMENTS IN SCREW GAUGE SCALE



Practice Worksheet

Test your skills by attempting these sample exam problems independently. Draw diagrams where appropriate:

Q1 (1-Mark): Differentiate between fundamental and derived physical quantities, giving three examples of each.

Q2 (2-Mark): State the rules for writing SI units and their symbols (capitalization, plurals, punctuation) as specified by BIPM.

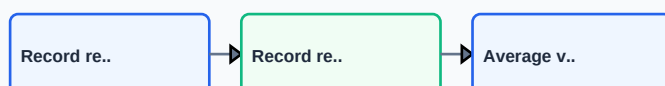
Q3 (2-Mark): A screw gauge has a PSR of 3 mm and HSC of 42. If it has a negative zero error of 5 divisions, find the thickness of the wire.

Q4 (5-Mark): What is the difference between physical balance and digital balance in terms of sensitivity and accuracy?

Q5 (5-Mark): Why is a degree sign omitted when writing temperatures in Kelvin (e.g., 273 K, not 273 °K)?

Q6 (5-Mark): A Vernier caliper has 10 divisions on the Vernier scale which coincide with 9 divisions on the main scale (1 division = 1 mm). If the jaws are closed and the Vernier zero is to the left of the main scale zero by 2 divisions, calculate the zero error and zero correction.

PRACTICE: PHYSICAL BALANCE MEASUREMENT SHEET



Real-World Applications & Connections

This chapter connects directly with modern industrial engineering, computation, and natural systems in numerous ways:

Application Case: Scientific Laboratories

Vernier calipers and screw gauges are standard instruments to measure the diameters of spherical lenses, capillaries, and wires with high precision.

Application Case: Aviation and Space Navigation

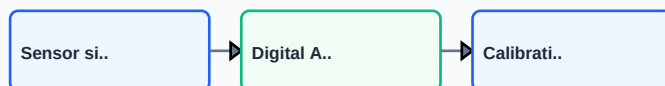
Astronomical units and light years are used to coordinate orbits, rocket vectors, and deep space telemetry without rounding errors.

Application Case: Metallurgical and Wire Manufacturing Industry

Used to gauge wire thicknesses, sheet metal profiles, and precision components in machining and aerospace fabrication to ensure tolerances.

Cross-Curricular Link: Concepts learned in this unit are heavily applied in other subjects like Physics (equations of motion, field vectors) and Data Analytics.

APPLICATION: INDUSTRIAL CALIBRATION IN ELECTRONICS



Revision Summary & Strategy

Use this checklist to self-evaluate your exam preparation status before final tests:

Key Recall Tips:

- ✓ Always write down 'Given Parameters' first to ensure partial marks are locked in.
- ✓ Pay attention to signs (+ / -) when performing algebraic operations or force mappings.
- ✓ Check that you have written the correct units (e.g. Joules, Tesla, cm^3 , N) for the final numeric answer.
- ✓ Draw neat diagrams using a sharp pencil and ruler for geometry or circuit layouts.

Syllabus Checklist:

- ☐ Read and memorized core definitions on Page 2.
- ☐ Reviewed critical formulas and relations on Page 3.
- ☐ Solved worked examples on Page 4.
- ☐ Attempted worksheet problems on Page 6.

CONGRATULATIONS! YOU HAVE COMPLETED THIS SYLLABUS STUDY GUIDE!

Practice consistently, verify answers with standard textbook methods, and take the topic test in the Nexora Learning app to gauge progress.
Good luck!

UNIT PREFIXES AND CONVERSIONS SCALE

